Did the provider run the model it billed you for?

MACHINE ALIGNMENT
TRANSPARENCY & SECURITY

Two channels for inference verification — the tokens it returned, and the clock — and what a verdict costs in each

Jiawei Li[1] · mentored by Gabriel Kulp[2] and Roy Rinberg[2]

[1]MATS 10.0 Scholar [2]MATS 10.0 Mentor Code + every run artifact: github.com/lijiawei20161002/inference-verification

MOTIVATION

Every compute-governance proposal assumes a claim nobody can currently check.

Model registration, compute thresholds, third-party audits, export-control attestations, EU AI Act GPAI duties — each presumes a statement of the form "model M ran under specification φ." Nothing in the API stack between a client and a GPU produces that statement. The client sees tokens; the provider sees the weights; a regulator inherits whichever it chooses to believe.

THE BUSINESS OF CHEATING

You rent inference. The provider commits to M under a sampling spec φ and bills per token. A 4-bit copy, a smaller sibling, or a truncated context is pure margin — and invisible in the tokens the client is paying for. But the deviations worth doing are exactly the ones that read fewer bytes, which is why they leave a second trace.

what the provider does instead	bytes it saves	worth it?
Serve Qwen3-0.6B as Qwen3-1.7B	2.83×	capability
4-bit (NF4) weights instead of bf16	2.55×	capacity
Attend half the contracted context	1.35×	capacity
Wrong seed / wrong temperature	1.00×	no

Table 1 Weight + KV bytes per decode step, computed from the timing grid's own byte counts (clock_channel.json, H100 PCIe, B=1). The last row is the control: a deviation that saves no bytes has no clock signature at all, and the poster's two channels split exactly there.

CONTRIBUTIONS

1. **One protocol, one grid.**
   8 detectors × 6 deviations on real models under a single standardized pAUC @ FPR ≤ 0.5% with an enforced batch/pool ceiling. Re-measured inside it, 16 of 24 previously published cells fall.
2. **Price the verdict, not the detector.**
   $(\delta^*/d')^2$ tokens × seconds per token. Detector rankings and FLOP counts are not what a client buys; 14 of 35 cells have no finite price at all.
3. **A second, orthogonal channel — and the honest version of it.**
   Absolute latency tests do not survive measurement; a differential clock test does, at 36–152 tokens of stream against the token channel's 2,013–35,066.

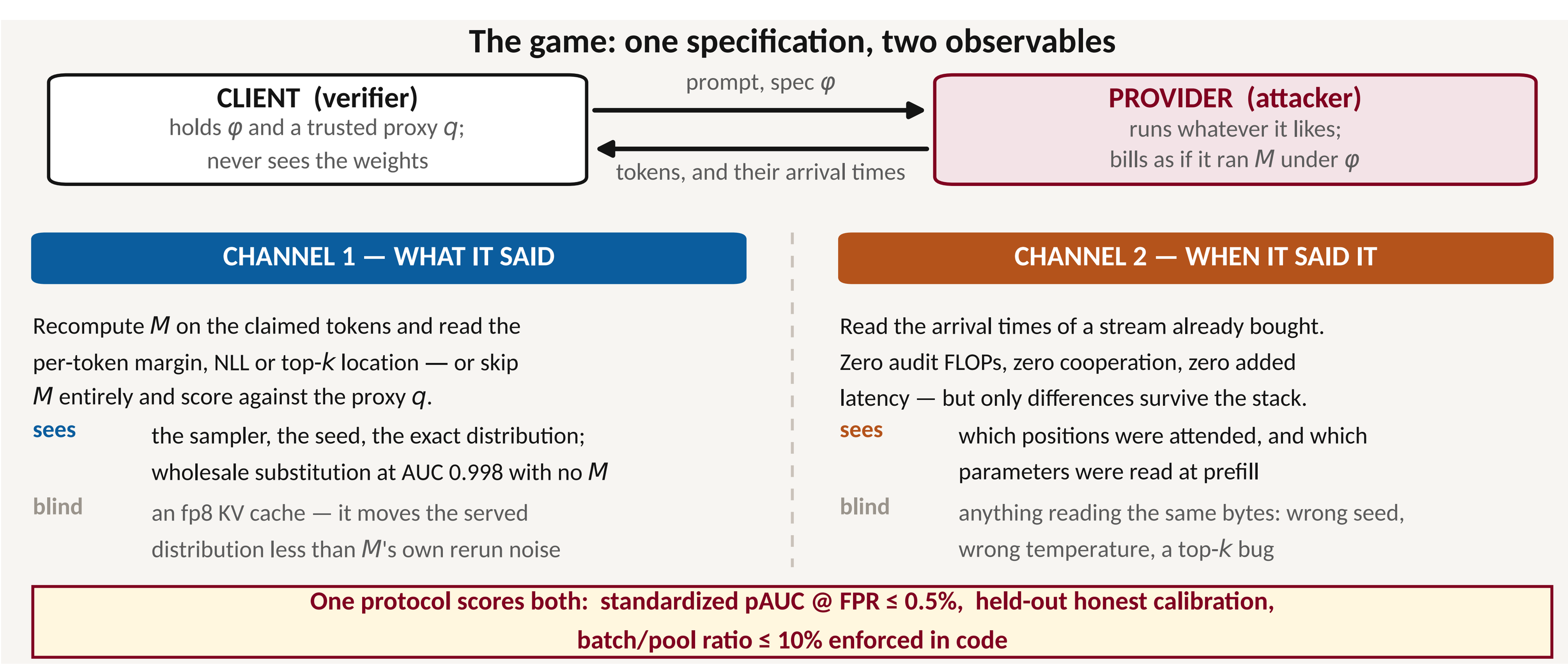

Figure 1 The two channels are complementary by mechanism, not by accident. A deviation either changes the distribution the tokens were drawn from, or it changes the bytes the GPU had to move, or both — and the two rows with the most money in them fall on opposite sides of that line.

CHANNEL 1 · THE TOKEN VERIFIER

One token carries almost nothing. The verdict is bought in bulk, and d' sets the price.

Re-running M and comparing per-token sampling margins catches every deviation tried here — on some model. But the per-token effect size from realistic quantization is $d' \approx 0.08$, so a batch of b tokens separates by only $d'\sqrt{b}$. That one number, measurable before any budget is committed, predicts the detection AUC to within 0.012.

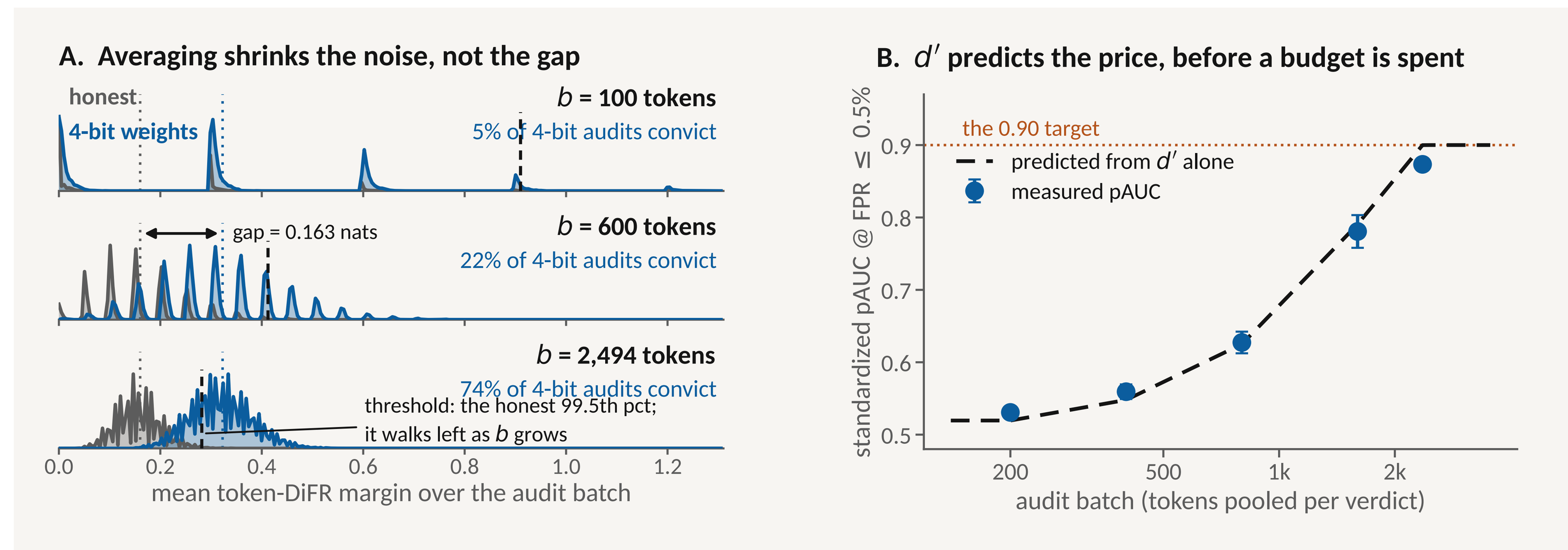

Figure 2 (A) 12,000 bootstrap audits per row from the experiment's own per-token scores (cost_of_a_verdict_scores.npz; Qwen3-1.7B, 80×256 tokens per arm, per-token d' = 0.075). The gap between the two means never moves; only the sampling noise does. (B) The same law tested on a 56,160-token pool (pool_scaling.json): five points inside the 10% ceiling, mean absolute residual 0.012.

WHAT CATCHES WHAT

Standardized pAUC @ FPR ≤ 0.5%, Qwen3-0.6B, 22,464-token honest pool, batch 1,000, 5 protocol seeds — a legitimate 8.9% batch/pool ratio. Every detector is structurally blind to a different class, and which one wins is set by the attack, not the model: use a portfolio.

	token DiFR	cross entropy	activation DiFR	token top-loc
4-bit weights	0.718	0.505	1.000	0.600
fp8 KV cache	0.511	0.507	1.000	0.517
wrong temperature	0.516	0.823	0.504	0.661
wrong seed	1.000	0.524	0.575	0.507
top-k bug (k=2)	0.553	0.510	0.499	0.523
top-k bug (k=32)	0.712	0.605	0.499	0.686

Table 2 headline_ratio.json. Re-scored from the same per-token scores as the previously published 78%-ratio table, 16 of 24 cells fall, by a median of 0.137. Under a wrong seed the provider redraws from exactly the honest distribution, so cross-entropy and top-loc can see nothing in principle — yet at 78% they read 0.987 and 0.828. An over-ratio pool does not exaggerate a signal; it invents one.

THE PRICE OF A VERDICT

Tokens a verdict needs, $(\delta^*/d')^2$ at δ^* = 3.77, times what a token costs. Tier 1 recomputes M at 141.1 µs/token; the "cheap" proxy tier is only 1.05× cheaper per token, because both are one latency-bound batch-1 prefill.

deviation	token DiFR	cross entropy	token top-loc	activation DiFR	accept rate	surface stat
4-bit weights	2,494	5,757	2,013	1	31,350	12,994
fp8 KV cache	35,066	—	35,430	2	—	—
wrong temperature	—	5,372	13,739	—	90,749	12,959
wrong seed	226	1,678,019	280,666	1,220,994	—	—
top-k bug (k=32)	1,886	2,264	1,443	—	—	6,691

Table 3 cost_of_a_verdict.json, Qwen3-1.7B audited with a Qwen3-0.6B proxy. A dash is a cell with no price — d' that fails its own detector's honest-vs-honest control, so no budget reaches a verdict through it. Ranking by AUC hides all 14 — 0.50 and 0.55 look adjacent while their prices are ∞ and 12,994 tokens — and a cheap token is not a cheap verdict: the proxy is 1.05× cheaper per token on 4-bit weights and needs 13× more of them.

CHANNEL 2 · THE CLOCK VERIFIER

A token cannot arrive before its bytes have moved — so the saving and the evidence are one quantity.

A batch-1 decode step is memory-bound: the margin is bytes not read, and bytes not read is time not spent. Zero audit FLOPs, zero cooperation, evidence arriving with a stream already bought. Measured on 126 timing cells, the premise holds and the naive test does not.

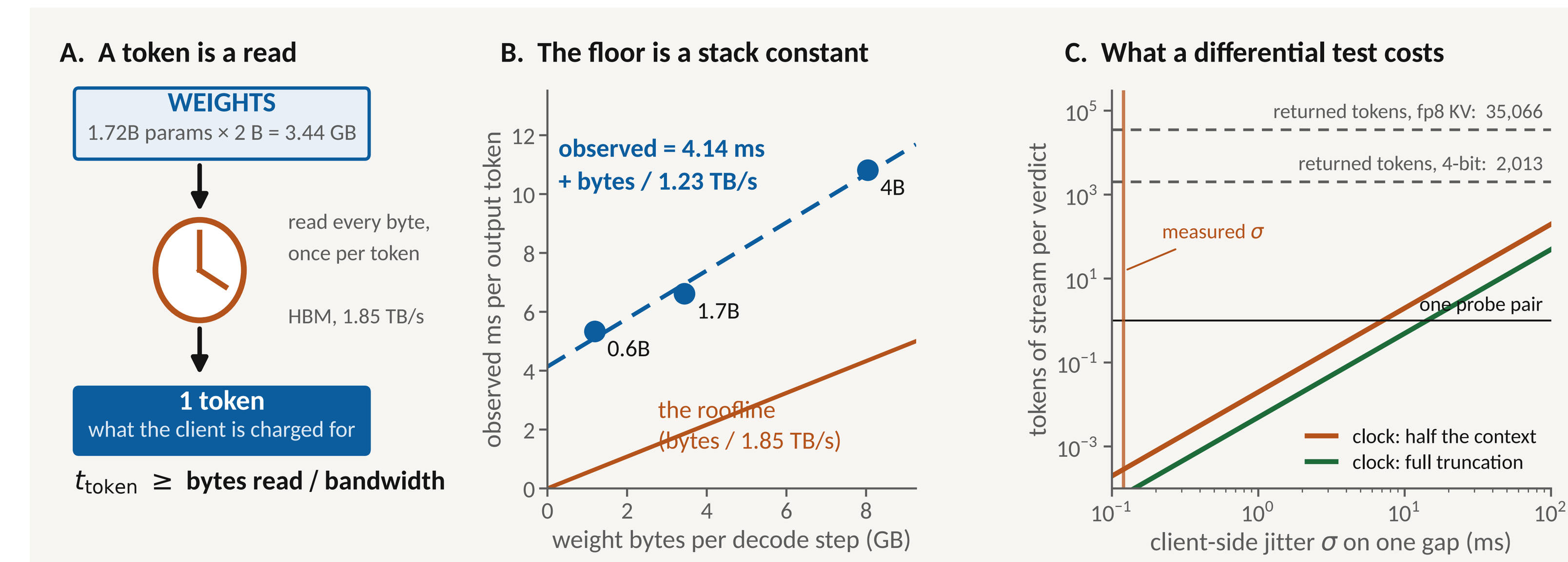

Figure 3 (B) clock_channel.json, H100 PCIe, CUDA-graph stack, B=1. The weight read runs at 66% of copy bandwidth but sits on a 4.14 ms constant that is pure stack and on no spec sheet — in eager HF the same GPU gives ~30 ms/token, flat in bytes: no channel at all. (C) Priced on the repo's own cost law.

WHAT MEASUREMENT AMENDED

the naive clock test says	measured
Fewer bytes → proportionally less time	NF4 delivers 14% of it
4-bit weights are 3.9× faster, so 6 tokens convict	1.10×, and slower at B≥4
Half the KV bytes → half the slope	int4 KV is 1.6× steeper
Jitter is positive, so take the minimum gap	73% of honest gaps are 0 ms
The floor is on the spec sheet	4.15 ms of it is the stack
One clock: the inter-token gap	prefill is 25,000× cheaper

Table 4 clock_channel.json (126 configurations) and clock_algos.json. The premise survives all six rows; the naive test survives none — and every failure attacks an absolute threshold, not a difference.

THE VERIFIER THAT SURVIVES

$D = \mathrm{ITL}(\mathrm{ctx}_{hi}) - \mathrm{ITL}(\mathrm{ctx}_{lo})$ over two matched probes: the stack constant, the network offset and any padding cancel exactly, and co-tenancy only raises the slope, so the honest floor is one-sided. A provider billing for 32,768 tokens and holding fewer, at 50 ms of wire jitter:

context it actually holds	d' / probe pair	pAUC @ 32	tokens
512 of 32,768	0.89	0.996	36
2,048	0.82	0.987	42
8,192	0.68	0.940	62
16,384 (half)	0.43	0.714	152

Table 5 slope_verifier.json, 42 cells through harness.evaluate on the house protocol. Re-measured an hour later in a fresh process the same cells drift by 0.01% in slope, so the test is self-calibrating and relative: it catches a provider that starts truncating. Evading it costs up to 53 ms per output token — the provider keeps its 3.7 GB of KV cache and gives back the speed.

LIMITATIONS

The wire is unmeasured: every jitter figure is device-side (sd 0.11–0.8 ms), a lower bound on a client's — hence the sweep in 3C. One H100, HF eager and CUDA graphs only; vLLM's paged attention may restore fp8-KV detectability. Models are 0.13B–8B, where the attacker's incentive is smallest, and quantization attacks are logit-level. No provider here adapts.

ACKNOWLEDGMENTS This work was carried out in the Machine Alignment, Transparency & Security (MATS) program, cohort 10.0. Thanks to Gabriel Kulp and Roy Rinberg for mentorship and to MATS for compute. Correspondence: jiawei2012220@gmail.com

REFERENCES

[1] Karvonen et al. DiFR: Inference Verification Despite Nondeterminism, 2025 — the Tier-1 methodology, reimplemented clean-room here. [2] RoyRin, inference_verification_for_model_weight_exfil — the threat model. [3] Every number on this board regenerates from docs/results/ via paper/make_mats_poster.py.

Code Paper